Evidence-Gated Terrain Admission for a Wheeled Delivery Robot

ECE 532 Final Project Proposal (graduate track, solo) Jeff Barner October 30, 2026
Repository: `https://github.com/nalddox/ml4ss-terrain-gate`

1 Problem and Decision Boundary

A wheeled delivery robot plans over a campus sidewalk graph whose edge labels come from a map prior (OpenStreetMap). Map labels can be stale or wrong; a shortest-path planner that trusts an edge mislabeled `footway` when it is actually stairs commits the robot to an untraversable, potentially damaging segment. **Decision:** for each edge e the planner proposes, may the robot commit to traversing e? The policy emits one of four outcomes: **Act** (admit and traverse); **Defer** (unresolved obligation and closer observations remain: continue approach and re-observe); **Repair** (an obligation fails, or remains unresolved with no evidence left: reject e, remove it, replan, and record a proposed map patch); **Handoff** (no admissible route to the goal remains: escalate to a remote operator). Costs are asymmetric: admitting stairs is catastrophic, a false rejection costs a detour. The gate is *fail-closed*: an edge is inadmissible unless evidence affirmatively discharges every obligation for its risk tier. **Research question:** can a calibrated, tiered multi-source gate match a conservative "reject if any source objects" veto on false admission while reducing unnecessary rejection and deferral, including under injected sensor and map faults?

2 Modalities, Data, Typed Claims, Risks, and Obligations

Data. *GND* [1] (CC0): wheeled-robot campus traversals with 3D LiDAR, RGB, GPS, IMU, and a human-labeled multi-category traversability map (including stairs) used as edge ground truth. The baseline uses *one* GND campus, subsampled to 2 Hz. *SANPO* [2] (CC BY 4.0): pixel-level stairs/curb segmentation, used only to train the terrain head, so SANPO→GND is a genuine viewpoint shift (eye height → robot height). *OSM* [3] (ODbL): footway graph for the GND campus, cached once as GeoJSON. Bags are converted once with the pure-Python `rosbags` library [4]; no ROS or network is needed afterward. Each observation is associated with an edge by projecting the edge polyline into the camera/LiDAR frame from the logged pose; ground-truth edge labels come from the GND traversability map at the edge's footprint.

Typed claims. Every claim is a typed record with fields `edge`, `predicate`, `value`, `uncertainty`, `source`, `tier`, and `provenance` (sensor, timestamp, frame index, SHA-256 of the cached artifact, and model/checker version).

Source	Example claim	Role
RGB + frozen embedding	`terrain(e)=stairs, p=0.91`	Calibrated semantic hazard evidence (A1/A2)
LiDAR geometry	`max_step(e)=0.14 m, n_pts=812`	Independent physical evidence (checker C1)
Map prior (OSM)	`tag(e)=footway`	Planning prior; assumed possibly stale
GPS + IMU	`on_edge(e), sigma=2.1 m, d_sep=3.4 m`	Attribution of evidence to one edge (checker C2)

Risk tiers. An edge's tier is the *maximum* hazard asserted by any source (escalation only; no source can lower a tier): **T0** flat path; **T1** curb, curb cut, or slope; **T2** stairs or step exceeding the platform limit h_{lim} (0.10 m, configurable).

Verification obligations (all must be PASS; any FAIL ⇒ Repair; any UNRESOLVED ⇒ Defer, or Repair if no closer evidence remains): **O1** (all tiers) attribution: C2 confirms the GPS uncertainty ellipse (2σ) intersects only e, which matters because ramps often run parallel to stairs. **O2** (T1+) geometry: C1 reports $h_{\max} < h_{\text{lim}}$ with point coverage $\geq n_{\min}$. **O3** (T2) corroboration: the calibrated $p(\text{stairs}) < \tau_2$ *and* O2 holds on ≥ 2 frames at distinct ranges. **O4** (all tiers) provenance: every claim used resolves to a hashed cached artifact. **Risks:** few real stair edges per campus (mitigated by Wilson CIs, per-observation analysis, and a second campus if needed); poor GPS near buildings (caught by O1); SANPO→GND shift degrading the camera (caught by O2/O3, and measured).

3 CPU Baseline: Estimated Runtime and Memory

Planning estimates on 4 cores / 16 GB, assuming ≤1 h of driving (≤7.2k frames at 2 Hz); all values will be measured and logged to `compute.json` at the progress gate. The backbone is a frozen DINOv2 ViT-S/14 [5] (fallback: MobileNetV3-Small) with a linear head.

Stage	Est. time	Peak RAM
One-time bag extraction and 2 Hz subsampling (streaming)	∼45 min	<2 GB
OSM graph, edge association, ground-truth edge labels	<5 min	<1 GB
Embed ∼6k SANPO + ∼7.2k GND frames (∼60 ms/frame)	∼15 min	∼3 GB
Head training and temperature calibration [6]	<3 min	<2 GB
C1 ground fit (RANSAC [8]) and step height (∼30 ms/frame)	∼5 min	<2 GB
Gate, planner, receipts: 4 policies × 9 conditions × 5 seeds	∼20 min	<3 GB
Full baseline / evaluator path from cache	∼1.5 h / ∼25 min	<6 GB

Both are well inside the 4 h baseline and 2 h evaluator budgets. Cached artifacts for evaluation (JPEG frames, downsampled LiDAR, embeddings, C1 outputs) are estimated at <8 GB. **Minimum fallback:** C1 + C2 + map prior with no learned model, which still exercises claims, tiers, gate, receipt, invariant, and failure injection end to end.

4 Checkers, Action Gate, Invariant, and Failure Injection

Two independent checkers use disjoint evidence and code paths. **C1, geometric step-height:** fits a local ground plane to LiDAR points on the edge footprint and reports the maximum surface discontinuity; it involves no learning and ignores semantic labels. **C2, localization attribution:** uses GPS/IMU covariance and graph geometry to decide whether an observation can be attributed to a single edge. **Gate:** a pure, deterministic function $(\text{claims}, \text{tier}) \mapsto \{\text{Act}, \text{Defer}, \text{Repair}, \text{Handoff}\}$. Missing evidence yields UNRESOLVED, never PASS. Deferral is real in replay: it consumes the next cached observation of e at closer range. **Invariant I1:** *no edge is admitted unless every obligation for its final tier is* PASS*, the tier is at least the maximum asserted by any claim, and every supporting claim's hash resolves.* I1 is checked on every receipt by `validate -strict` and by property-based tests over randomized claim sets. **Repair:** the planner removes the rejected edge and replans; the receipt records a proposed map patch (`tag(e): footway→steps`) together with the evidence that justified it. **Receipt:** `receipt.json` records config and data-manifest hashes, seed, injected faults, and per-edge claims, obligation states, decision, and A3 attribution, plus the I1 result.

Failure injection (each paired with its expected detection and declared in the receipt): *missing*: camera or LiDAR dropout (⇒ UNRESOLVED, never admit); *corrupted*: blur, low light, glare [9]; LiDAR thinning; GPS bias of 3–10 m and random-walk drift (⇒ O1 fails); *stale map*: flip k% of stair edges to `footway`; *shifted*: SANPO→GND viewpoint; *adversarial evidence substitution*: flip the map *and* replace stair image crops with flat-sidewalk crops, so that two of three sources agree on "flat" and only C1 stands between the robot and the stairs. **Explicit unwarranted claims:** no claim of real-world deployment safety; of generalization beyond the evaluated campus; of detecting hazards other than steps, stairs, and curbs (ice, potholes, pedestrians); of step heights below LiDAR resolution; or of physical causation. A3 attributions describe the decision procedure, not the world.

5 Metrics, Baselines, Ownership, Timeline, and Fallback

Task performance (reported separately): stairs AUROC/F1 and ECE on GND, and C1 agreement with ground-truth step edges. **Policy behavior:** false admission rate (primary; Wilson 95% CI), unnecessary rejection, unnecessary deferral, handoff rate, checker coverage (fraction of admitted edges whose obligations were all discharged by an executable checker), failure detection (fraction of injected faults flagged in the receipt), repair success (replanned route reaches the goal over ground-truth-traversable edges only; overhead relative to an oracle-map route), runtime, and peak RSS. **Baselines:** B0 map-only shortest path; B1 camera-only threshold; B2 conservative veto (reject if any source objects); **B3** the proposed calibrated tiered gate; plus an oracle-map route for reference. **A1/A2/A3:** consume cached embeddings as representation evidence and extend them with a SANPO→GND embedding-shift score that is itself a typed claim (A1); consume calibrated evidence and extend it to multi-source, per-tier thresholds that bound false admission on a calibration split [7] (A2); consume the typed causal claim and extend it to per-decision attribution ("e rejected *because of* C1"), verified exactly by deterministic counterfactual re-runs of the gate with each source ablated (A3). **Track ownership:** solo; I own all tracks (data/replay, perception, checkers/gate, evaluation/receipts).

Timeline. *Oct 31–Nov 6:* repository, lockfile, GND campus download and extraction, OSM cache, smoke config. *Nov 7–13:* C1, C2, receipt schema, B0/B2. *Nov 14–20:* embeddings, head, calibration, tiered gate and policy (B1/B3). *Nov 21–30:* injection harness, I1 property tests, measured runtime and memory. **Dec 1 progress gate:** all gate items. *Dec 2–10:* full evaluation, A3 attribution, poster (Dec 11). *Dec 12–15:* report. **Fallbacks:** if GND extraction or label alignment fails for one campus, switch campuses or hand-label ∼200 edges from the GND map; if the camera head underperforms, report it and rely on the minimum (no-model) fallback; if stair edges are too few, pool two campuses.

References (annotated)

[1] J. Liang et al., "GND: Global Navigation Dataset with multi-modal perception and multi-category traversability in outdoor campus environments," arXiv:2409.14262, 2024. *Primary replay data and ground truth (traversability map, including stairs).*

[2] S. M. Waghmare et al., "SANPO: A scene understanding, accessibility, navigation, pathfinding, obstacle avoidance dataset," arXiv:2309.12172, 2023. *Stairs/curb segmentation labels for training the terrain head; provides the viewpoint shift.*

[3] OpenStreetMap contributors, `https://www.openstreetmap.org`, ODbL. *Map prior (footway/steps tags) that the gate treats as possibly stale.*

[4] Ternaris, `rosbags` (pure-Python ROS bag I/O), `https://gitlab.com/ternaris/rosbags`. *One-time bag conversion with no ROS install.*

[5] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," TMLR, 2024. *Frozen compact backbone for cached embeddings (A1).*

[6] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, "On calibration of modern neural networks," ICML, 2017. *Temperature scaling and ECE for calibrated terrain confidence (A2).*

[7] A. N. Angelopoulos, S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," arXiv:2107.07511, 2021. *Distribution-free threshold selection to bound false admission per tier.*

[8] M. A. Fischler, R. C. Bolles, "Random sample consensus," CACM 24(6), 1981. *Robust ground-plane fitting in checker C1.*

[9] D. Hendrycks, T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," ICLR, 2019. *Corruption families and severities for camera failure injection.*

A Data-Access Confirmation, Repository, and Disclosures

GND: CC0 1.0, publicly downloadable. [TODO: confirm campus chosen, download completed, size on disk, and that the traversability map and sensor extrinsics are present.] **SANPO:** CC BY 4.0, public. [TODO: confirm subset downloaded.] **OSM:** ODbL; campus extract cached as GeoJSON at setup. No account, payment, or API is required. **Repository:** `https://github.com/nalddox/ml4ss-terrain-gate`. **Disclosures (pending instructor feedback on the idea brief):** (1) A self-recorded campus sequence is *not* part of the graded path; if permitted it will be released with the repository and used only as qualitative held-out evidence. (2) This project may share a frozen vision front-end with my Machine Vision course project; the ML4SS deliverables (claims, tiers, obligations, checkers, gate, policy, receipts, failure-injection evaluation) are distinct and developed only here.

B Proposed Configurations

`config/smoke.yaml` (target <2 min)

```
seed: 0
device: cpu
data:
  campus: gnd_TODO
  max_frames: 200
  rate_hz: 2
  cache_dir: data/cache
graph: {max_edges: 20}
perception:
  backbone: dinov2_vits14
  embeddings: cached
  head: linear
checkers:
  c1_step: {h_lim_m: 0.10, n_min: 200}
  c2_loc: {k_sigma: 2.0}
gate: {threshold_method: fixed,
       tau_t2: 0.05}
policies: [b2_veto, b3_tiered]
injections: [gps_bias_5m]
seeds: [0]
out: experiments/smoke
```

`config/cpu.yaml` (target <2 h from cache)

```
seed: 0
device: cpu
data:
  campus: gnd_TODO
  max_frames: null
  rate_hz: 2
  cache_dir: data/cache
graph: {max_edges: null}
perception:
  backbone: dinov2_vits14
  embeddings: cached
  head: linear
  calibration: temperature
checkers:
  c1_step: {h_lim_m: 0.10, n_min: 200}
  c2_loc: {k_sigma: 2.0}
gate:
  threshold_method: conformal
  false_admission_target: 0.01
policies: [b0_map, b1_camera,
           b2_veto, b3_tiered]
injections: [none, cam_dropout,
  lidar_dropout, blur, low_light,
  gps_bias, gps_drift, map_flip,
  substitution]
seeds: [0, 1, 2, 3, 4]
out: experiments/cpu
```

C Optional Scale-Up Justification (separately labeled; not on the graded path)

No scale-up is required, and every graded result comes from the CPU baseline. If central resources are awarded after the progress gate, I would request $\leq$4 GPU hours to fine-tune the terrain head end to end on SANPO, to test whether reduced viewpoint shift lowers unnecessary rejection at equal false admission. Outputs would be cached as frozen embeddings and replayed deterministically, and GPU usage would be reported separately. No paid API is requested.

D Reproducibility Plan (preview of the graduate appendix)

```
pytest -q
python -m src.run --config config/smoke.yaml --device cpu
python -m src.evaluate --config config/cpu.yaml --device cpu
python -m utilities.verification.validate --strict experiments/receipt.json
```

The pipeline uses a pinned environment lockfile, fixed seeds, SHA-256 data manifests, cached embeddings and C1 outputs, and no network after setup.